

Application

- IEEE 802.11g
- IEEE 802.11b
- IEEE 802.11n

Features

- Easy evaluation of SiGe SE2576L
- Complete design - ready for production
- Integrates: PA, matching and power detector

Product Description

The SE2576L-EK1 Evaluation Kit is a tool that is intended to demonstrate the performance of the SE2576L 802.11bgn Power Amplifier in an 802.11bgn application. The kit facilitates and is optimized for evaluation, experimentation and investigation with an 802.11bgn signal source. The SE2576L-EK1 design and layout can be quickly and easily transferred into a production design.

The SE2576L-EK1 Evaluation Kit is comprised of a SE2576L evaluation board, test report, a SE2576L-EK1 datasheet and a SE2576L datasheet.

Order Information

Type	Package	Remark
SE2576L	16 Pin QFN	Samples
SE2576L-R	16 Pin QFN	Tape and Reel
SE2576L-EK1	Evaluation Kit	Standard

Functional Block Diagram

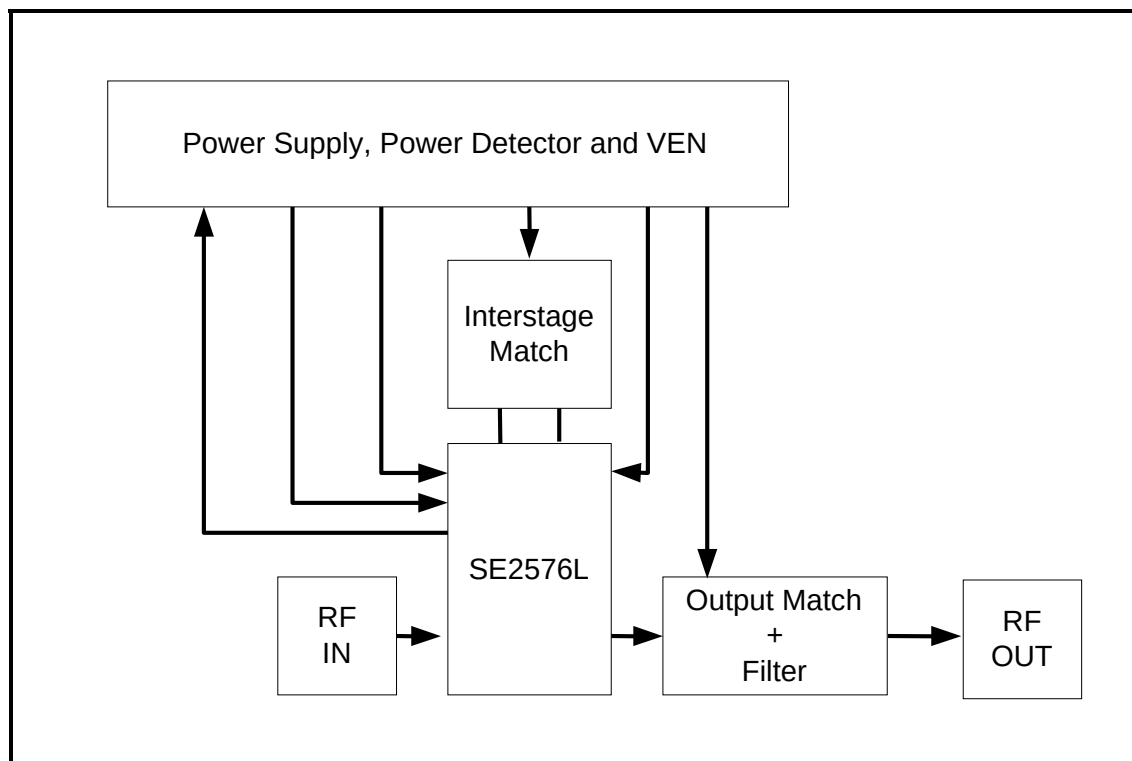


Figure 1: SE2576L Evaluation Board Block Diagram

Evaluation Board Layout

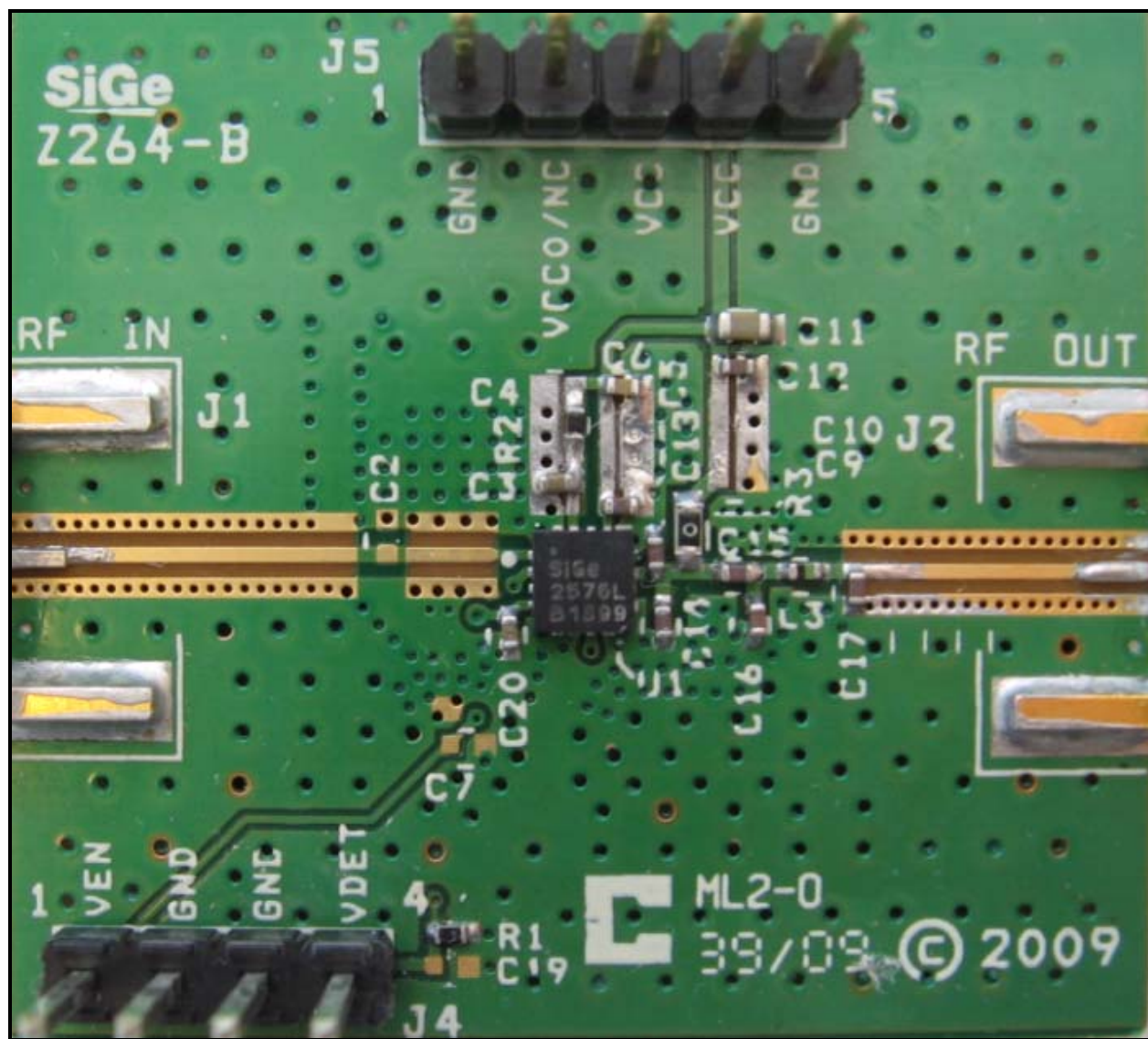


Figure 2: SE2576L Evaluation Board Layout Top Layer

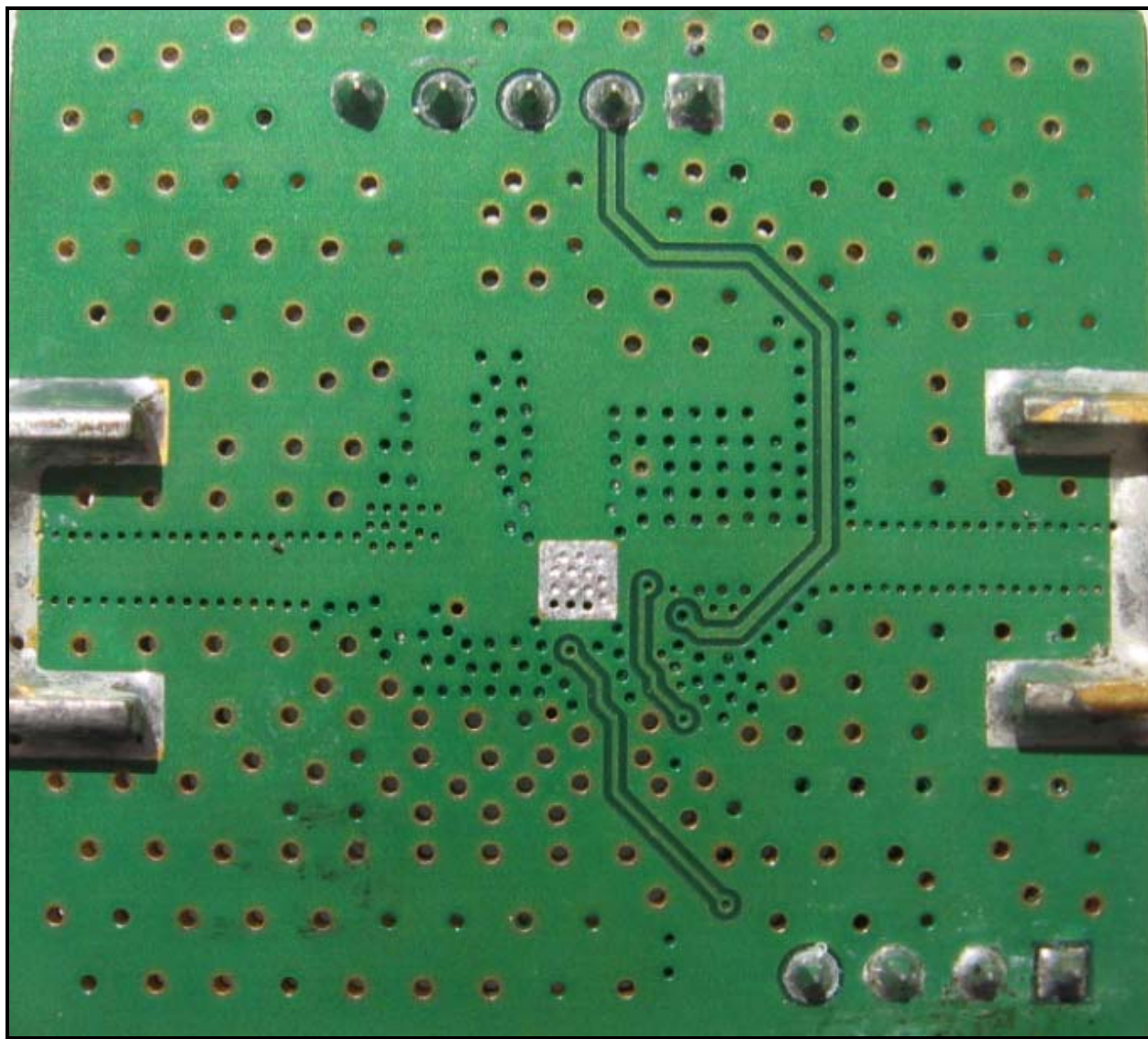


Figure 3: SE2576L Evaluation Board Layout Bottom Layer

Contents of Evaluation Kit

Quantity	Description	Reference
1	SE2576L Evaluation Board	SE2576L-EV1
1	SE2576L Datasheet	DST-00260
1	SE2576L Evaluation Kit Datasheet	DST-00287

Table 1: Contents of Evaluation Kit

Evaluation Board Schematic

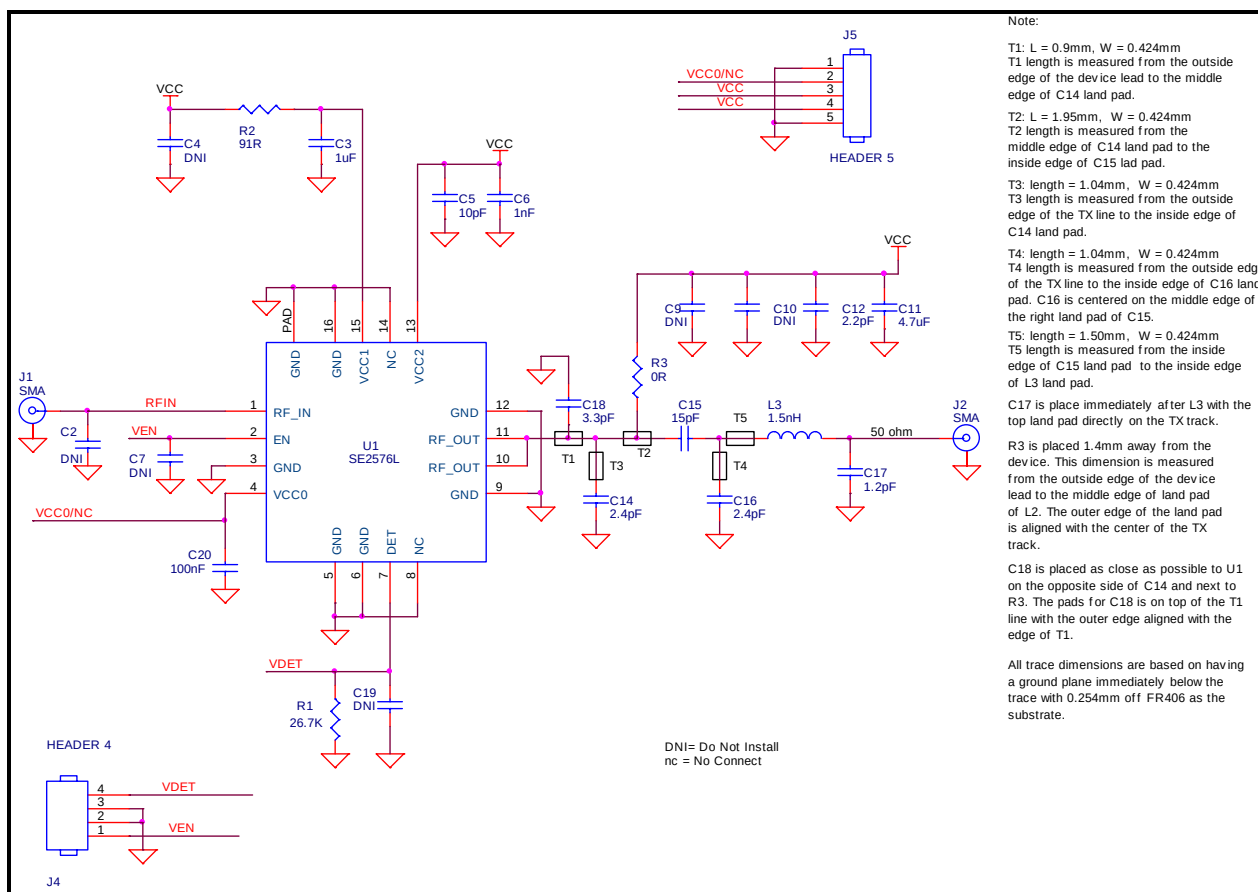


Figure 4: Evaluation Board Schematic Diagram

Getting Started

This section provides the details required to setup the evaluation board and the test equipment for the SE2576L-EK1 evaluation kit. Table 2 describes the pins on the Power and Analog I/O Header. Note: It is recommended to use proper engineering connection practices by making RF and digital connections prior to turning on the power supply.

1. Connect Supply.
 - a. Connect the supply ground to pin 1 of J5 and pin 2 of J4.
 - b. Connect 5 V to pins 2, 3 and 4 of J5.
2. Power Amplifier Enable. By applying 3.3 V on VEN (Pin 1, J4), the power amplifier is enabled. By placing a GND on VEN (Pin 1, J4), the power amplifier is disabled and placed in a “shutdown” state, drawing minimal current.
3. Measure Performance.
 - a. 2.5GHz amplifier performance can be monitored by applying an RF signal to connector J1 (RFIN), and monitoring the output power on the RFout port connector J2.
 - b. Detector performance can be monitored on pin 4 of J4.
 - c. Care should be taken not to overdrive the amplifier by applying too much RF on the input to the device. -20dBm provides a suitable starting input power for the device.

Pin	Pin Number	Description	Recommended setting
J5 Header 5 x 1			
VCC0/VCC	2,3,4	Supply Voltage	Connect to 5 V power supply
GND	1,5	Ground	Connect to Ground
J4 Header 4 x 1			
VEN	1	Power amplifier enable	Enable: Connect 3.3 V Disable: Connect to ground
GND	2,3	Ground	Connect to Ground
VDET	4	Detector output	Monitor voltage by connecting to volt-meter

Table 2: Power and Analog I/O Headers

Evaluation Board Bill of Materials

Table 3: Shows the components used in the design of the SE2576L evaluation board. SiGe Semiconductor does not endorse any specific vendors.

Item	Quantity	Reference	Part	Manufacturer	Manufacturer Part Number
1	6	R1,C2,C4,C9,C10,C19	DNI	-	-
2	1	C3	1uF	MURATA	GRM155R60J105KE19
3	1	C5	10pF	MURATA	GRM1555C1H100JZ01
4	1	C6	1nF	Murata	GRM155R71H102KA01
5	1	C11	4.7uF	Murata	GRM188R60J475KE19
6	1	C12	2.2pF	Murata	GRM1555C1H2R2CZ01
7	2	C14,C16	2.4pF	Murata	GRM1555C1H2R4CZ01
8	1	C15	15pF	MURATA	GRM1555C1H150JZ01
9	1	C17	1.2pF	MURATA	GRM1555C1H1R2CZ01
10	1	C18	3.3pF	MURATA	GRM1555C1H3R3CZ01
11	1	C20	100nF	MURATA	GRM155R61A104KA01
12	2	J1,J2	SMA	Johnson	142-0701-851
13	1	J3	HEADER 7	Samtec	TSW-107-07-G-S
14	1	L3	1.5nH	MURATA	LQG15HN1N5S02D
15	1	PCB	Z264-B	SiGe Semiconductor	Z264-B
16	1	R2	91R	Panasonic	ERJ2GEJ910
17	1	R3	0R	Panasonic	ERJ2GEJ0R0
18	1	U1	SE2576L	SiGe Semiconductor	SE2576L

Table 2: SE2576L Evaluation Board BOM

Notes:

- Schematic and BOM has been designed to optimize performance in 802.11bgn applications.

Document Change History

Revision	Date	Notes
1.0	Mar-2-2009	Created
1.1	Apr-22-2009	Updated schematic and BOM
1.2	Jun-29-2009	Updated schematic, BOM, and layout information
1.3	Apr-23-2010	Updated schematic, BOM, Layout and Getting Started section

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Product Preview

The datasheet contains information from the product concept specification. SiGe Semiconductor, Inc. reserves the right to change information at any time without notification.

Preliminary Information

The datasheet contains information from the design target specification. SiGe Semiconductor, Inc. reserves the right to change information at any time without notification.

Production testing may not include testing of all parameters.

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